R21

Code No.: U21PC602CS

MATURI VENKATA SUBBA RAO (MVSR) ENGINEERING COLLEGE

(Autonomous)

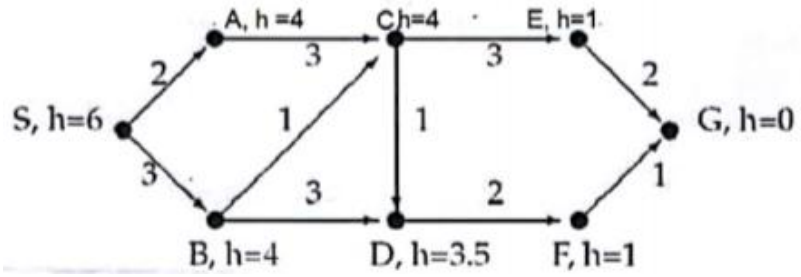
B.E. VI Semester (Supple.) (Branch: CSE) Examination, March 2025

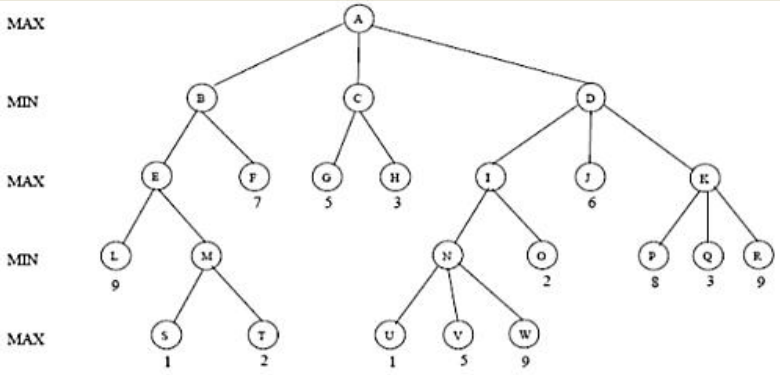
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 hours Max. Marks : 70

Note :
i) FIRST Question is compulsory and answer any **FOUR** questions from the remaining six questions. Each question carries 14 Marks.
ii) Answers to each question must be written at one place only and in the same order as they occur in the question paper.
iii) Missing data, if any, may suitably be assumed.

Q.No.		Marks	CO	BT
1. a)	Name the four basic kinds of agent programs.	(2)	1	L1
b)	List out the applications of expert systems.	(2)	2	L1
c)	Differentiate supervised and unsupervised learning.	(2)	3	L2
d)	Define perceptron.	(2)	4	L1
e)	Where do we use natural language processing?	(2)	5	L2
f)	What is perfect decision game?	(2)	1	L1
g)	How to measure the similarity between two objects?	(2)	5	L2

2	(a)	Explain the concept of states and operators, and how they are used to define the state space in problem-solving.	(7)	1	L3
	(b)	Outline the steps of A* algorithm and find the optimal path for the following graph using A* algorithm. 	(7)	1	L3
3	(a)	Explain the expert system architecture and its components.	(7)	2	L2
	(b)	Discuss the syntax and semantics of propositional logic.	(7)	2	L4
4	(a)	Illustrate the role of gradient descent method for linear regression.	(7)	3	L2
	(b)	Analyze the methodology used in Naïve Bayes classifier.	(7)	3	L4
5	(a)	Elaborate the approach of Back propagation algorithm.	(7)	4	L2
	(b)	How to use the accuracy, precision and recall for model evaluation? Discuss.	(7)	4	L3
6	(a)	Elucidate the strategy of K-Means algorithm with suitable example.	(7)	5	L2
	(b)	Discuss the applications of natural language processing.	(7)	5	L4
7	(a)	Apply alpha-beta pruning for finding the optimal path	(5)	1	L3

					
	(b)	What are the different metrics used in deciding the splitting attribute in decision trees?	(5)	3	L1
	(c)	Assess the functionality of various activation functions in neural networks.	(4)	5	L3

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